HSE 103 Final Project - Gene Expression

Research question

How does gene expression for CXCR2, CYP24A1, and PTX3 vary with maternal vitamin D concentration and pregnancy outcome?

The continuous covariate is maternal vitamin D (ng/mL). The two categorical variables are pregnancy outcome and vitamin D group (<30 versus >=30 ng/mL).

Import and prepare the original data

```
# Read the three course CSV files without editing them.
# These files need to be in the same folder as this R Markdown file.
library(tidyverse)
```

```
## Warning: package 'tidyr' was built under R version 4.5.2
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
covariates <- read.csv("GSE85307_covar-1.csv")
expression <- read.csv("GSE85307_geneExpression.csv", row.names = 1)
linker <- read.csv("GSE85307_geneNames.csv")

# The expression file has genes in rows and samples in columns.
# Transpose it so sample IDs become rows and probe IDs become columns.
expression <- t(expression)

# Inspect sample IDs before combining the datasets.
# In the RStudio Console, compare these IDs if you need to check ordering:
# head(covariates$geoID)
# head(rownames(expression))

# Clinical columns contain a description before the colon.
# Separate the description from each recorded value.
covariates <- separate(covariates, mat_vitD,
                       into = c(NA, "vitamin_d"), sep = ": ")
covariates <- separate(covariates, preg_preeclampsia,
                       into = c(NA, "outcome"), sep = ": ")

# Set the types used in the plots.
covariates$vitamin_d <- as.numeric(covariates$vitamin_d)
covariates$outcome <- factor(covariates$outcome,
                             levels = c("Normal", "Preeclampsia"))
covariates$vitamin_d_group <- factor(
```

```
  ifelse(covariates$vitamin_d < 30, "Below 30", "30 or higher"),
  levels = c("Below 30", "30 or higher"))

# Start one data frame with one row per clinical sample.
analysis <- data.frame(
  geoID = covariates$geoID,
  vitamin_d = covariates$vitamin_d,
  outcome = covariates$outcome,
  vitamin_d_group = covariates$vitamin_d_group
)

# Find each gene's probe ID in the linker file.
# Select expression values using the sample IDs, not original row positions.
gene_names <- c("CXCR2", "CYP24A1", "PTX3")
for (gene in gene_names) {
  probe <- as.character(linker$PROBEID[linker$GENE == gene])
  analysis[, gene] <- expression[covariates$geoID, probe]
}

# Inspect the analysis data frame before plotting.
# In the RStudio Console, inspect the prepared data before plotting:
# head(analysis)
```

Function to generate three plots for any selected gene

The function takes two inputs: a data frame and a vector of one or more gene names. It uses a `for` loop to make a histogram, scatterplot, and grouped boxplot for each gene.

```
make_gene_plots <- function(data, gene_names) {
  # Repeat the same three plot types for each selected gene.
  for (gene in gene_names) {

    # Put the selected gene expression and covariates into one plot data frame.
    plotData <- data.frame(
      Expression = data[, gene],
      VitaminD = data$vitamin_d,
      Outcome = data$outcome,
      VitaminDGroup = data$vitamin_d_group
    )

    # Histogram: distribution of expression for this gene.
    histogram <- ggplot(plotData, aes(x = Expression)) +
      geom_histogram(binwidth = 0.10, fill = "steelblue", color = "white") +
      labs(title = paste(gene, "expression distribution"),
           x = "Gene expression (log2 signal)",
           y = "Number of participants") +
      theme_classic()

    # Scatterplot: vitamin D on the x-axis and expression on the y-axis.
    scatterplot <- ggplot(plotData, aes(x = VitaminD, y = Expression)) +
      geom_point(color = "steelblue", alpha = 0.7) +
      geom_smooth(method = loess) +
```

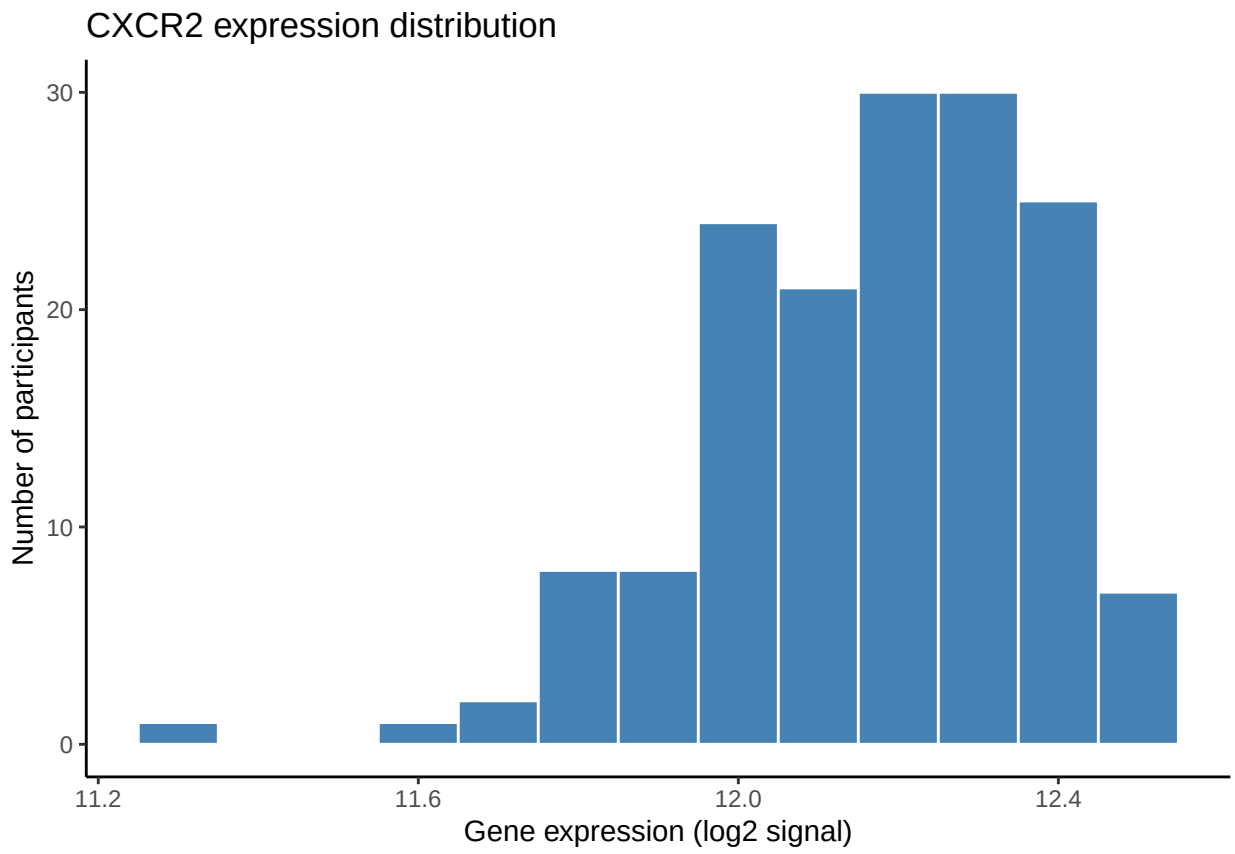
```
      labs(title = paste(gene, "expression and maternal vitamin D"),
           x = "Maternal vitamin D (ng/mL)",
           y = "Gene expression (log2 signal)") +
      theme_classic()

    # Boxplot: pregnancy outcome separated by the two vitamin D groups.
    boxplot <- ggplot(plotData, aes(x = Outcome, y = Expression,
                                    fill = VitaminDGroup)) +
      geom_boxplot(position = position_dodge(0.9)) +
      scale_fill_manual(values = c("steelblue", "orange")) +
      labs(title = paste(gene, "expression by pregnancy outcome"),
           x = "Pregnancy outcome",
           y = "Gene expression (log2 signal)",
           fill = "Vitamin D (ng/mL)") +
      theme_classic()

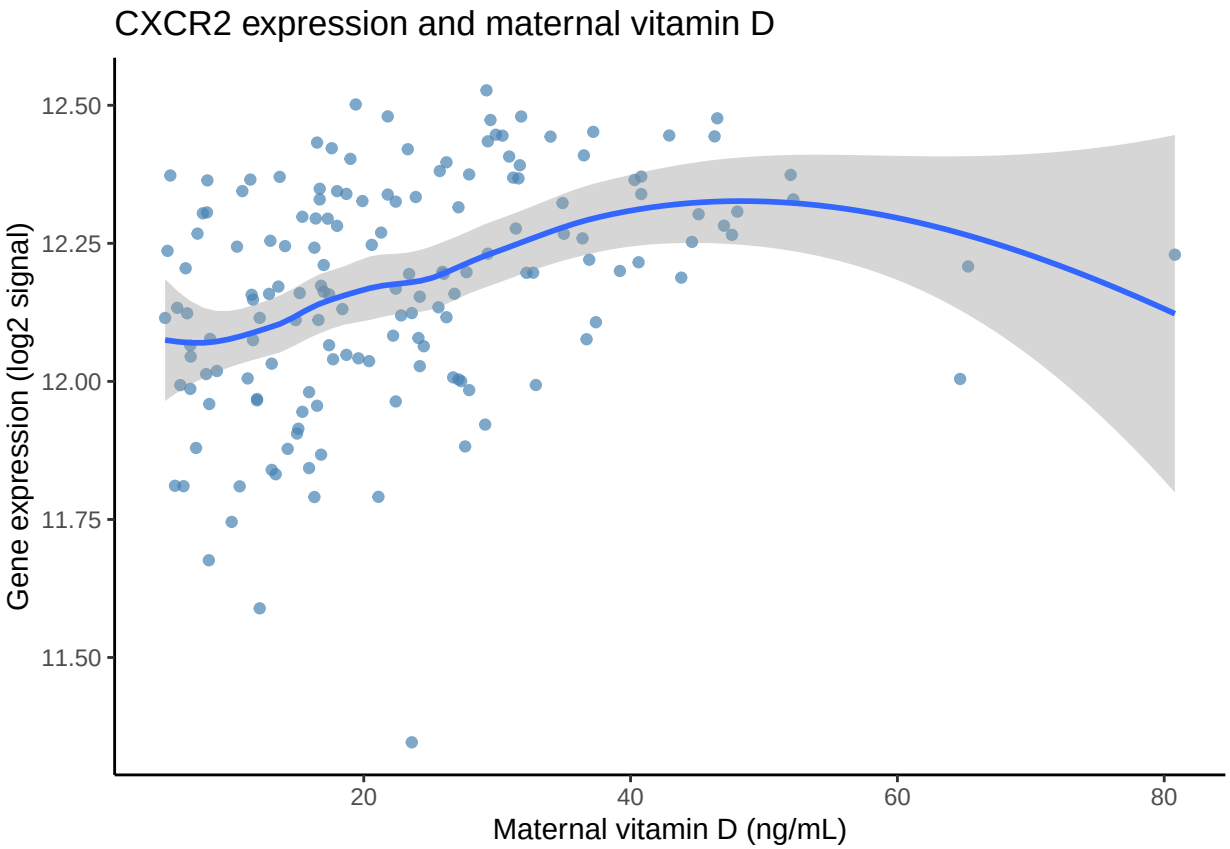
    # print() displays plots made inside a for loop in the knitted report.
    print(histogram)
    print(scatterplot)
    print(boxplot)
  }
}
```

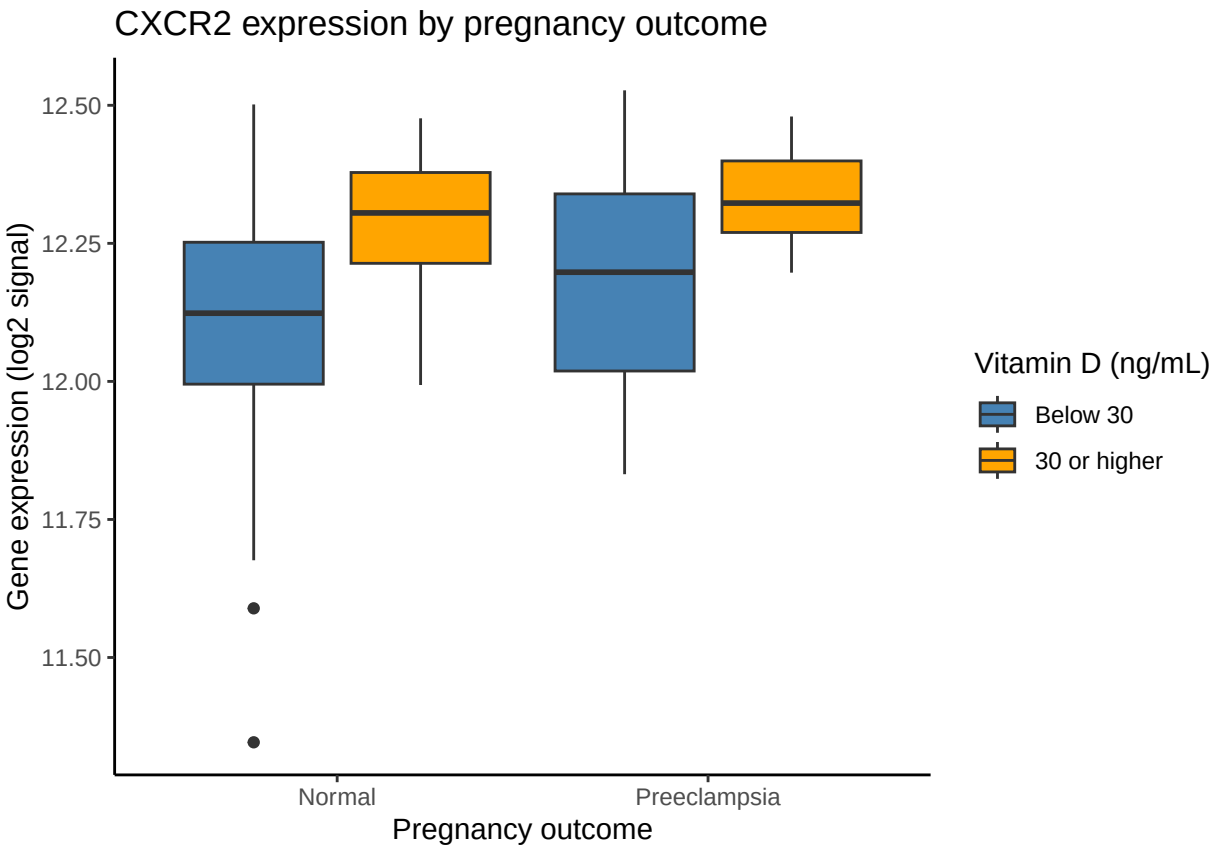
Nine plots for three genes

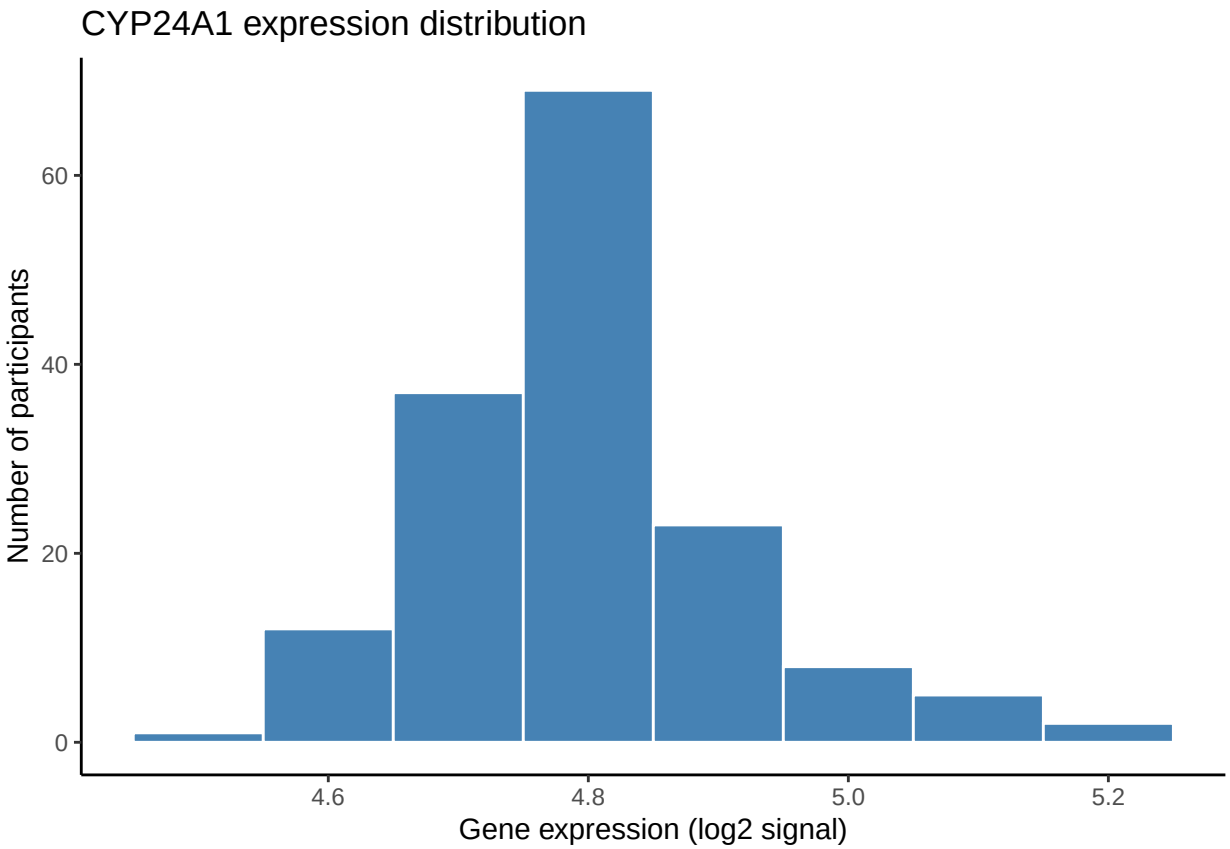
```
# Three genes, each with one histogram, one scatterplot, and one boxplot.
make_gene_plots(analysis, gene_names)
```



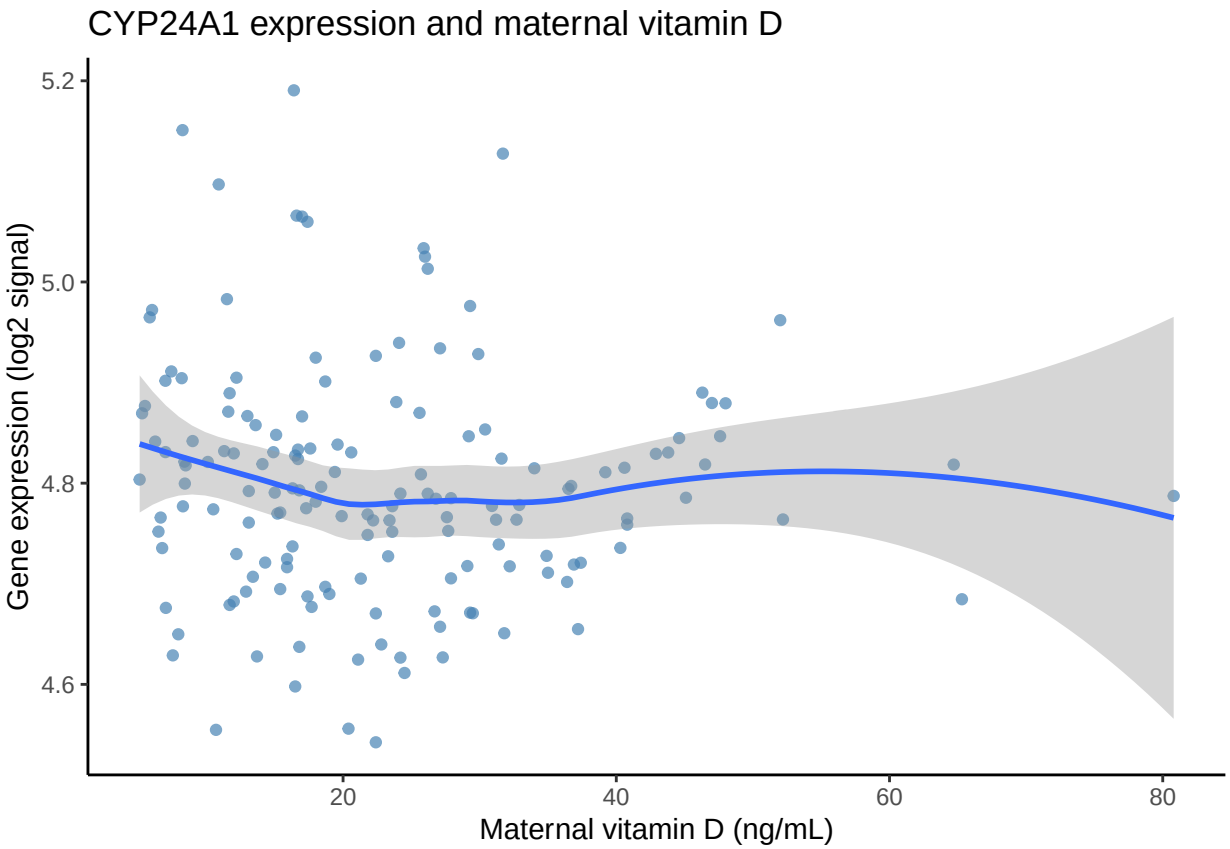

```
## `geom_smooth()` using formula = 'y ~ x'
```

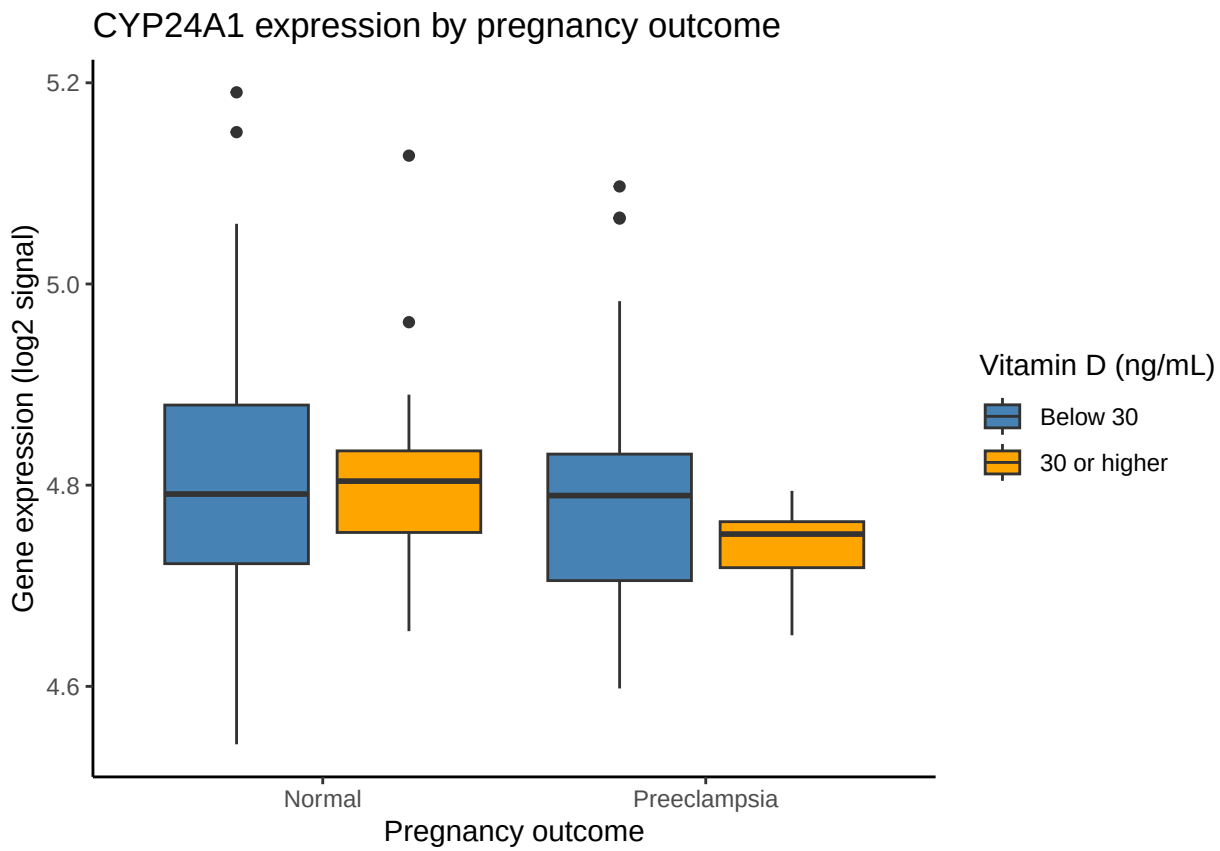

CXCR2 expression and maternal vitamin D
Gene expression (log2 signal)
12.50
12.25
12.00
11.75
11.50
20
40
60
80
Maternal vitamin D (ng/mL)

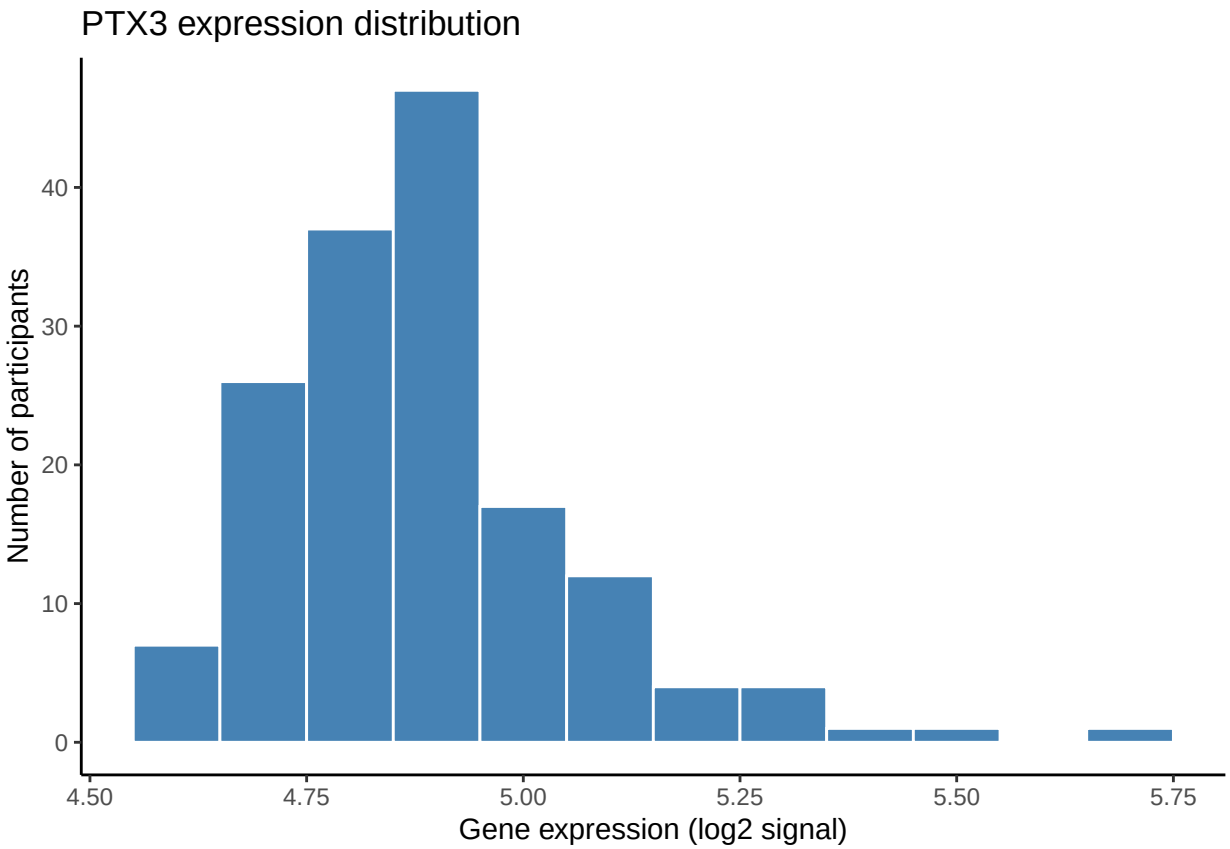

CXCR2 expression by pregnancy outcome
Gene expression (log2 signal)
12.50
12.25
12.00
11.75
11.50
Normal
Preeclampsia
Pregnancy outcome
Vitamin D (ng/mL)
Below 30
30 or higher



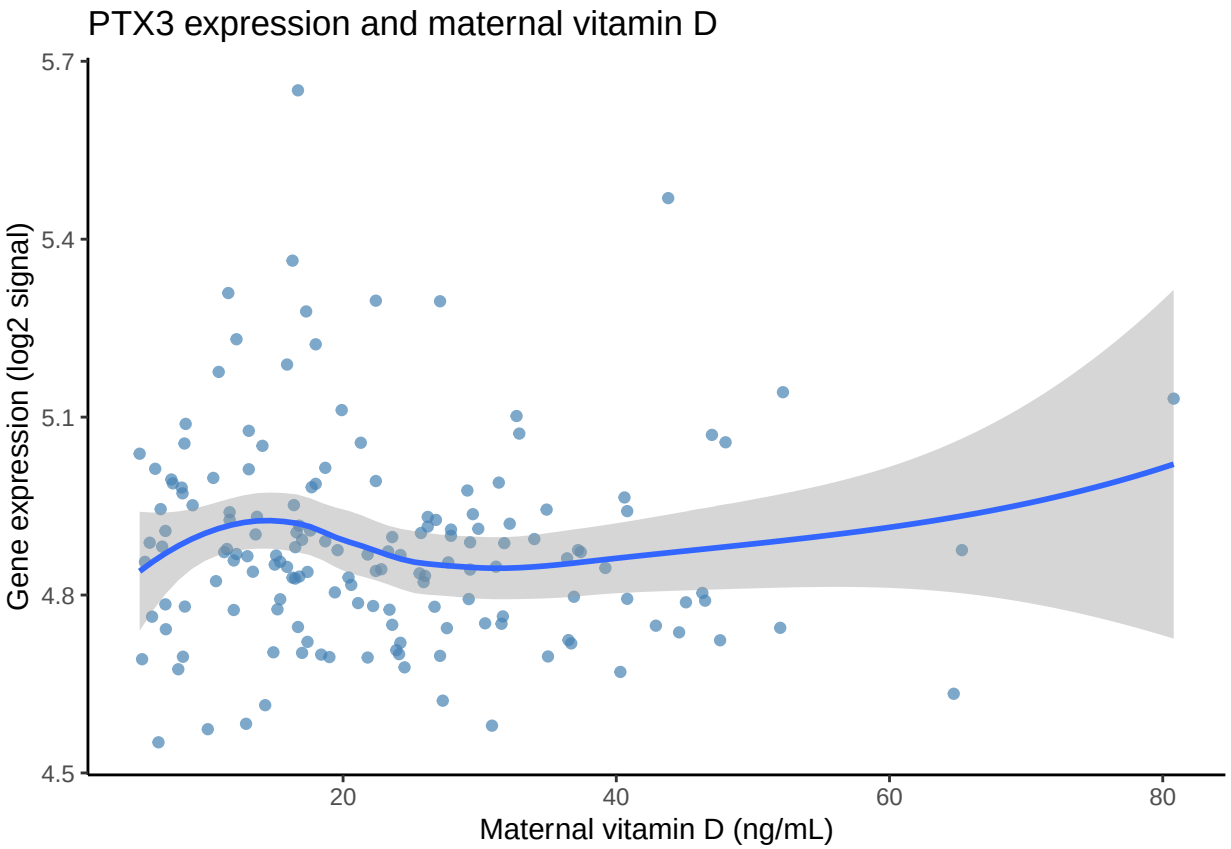

```
## `geom_smooth()` using formula = 'y ~ x'
```

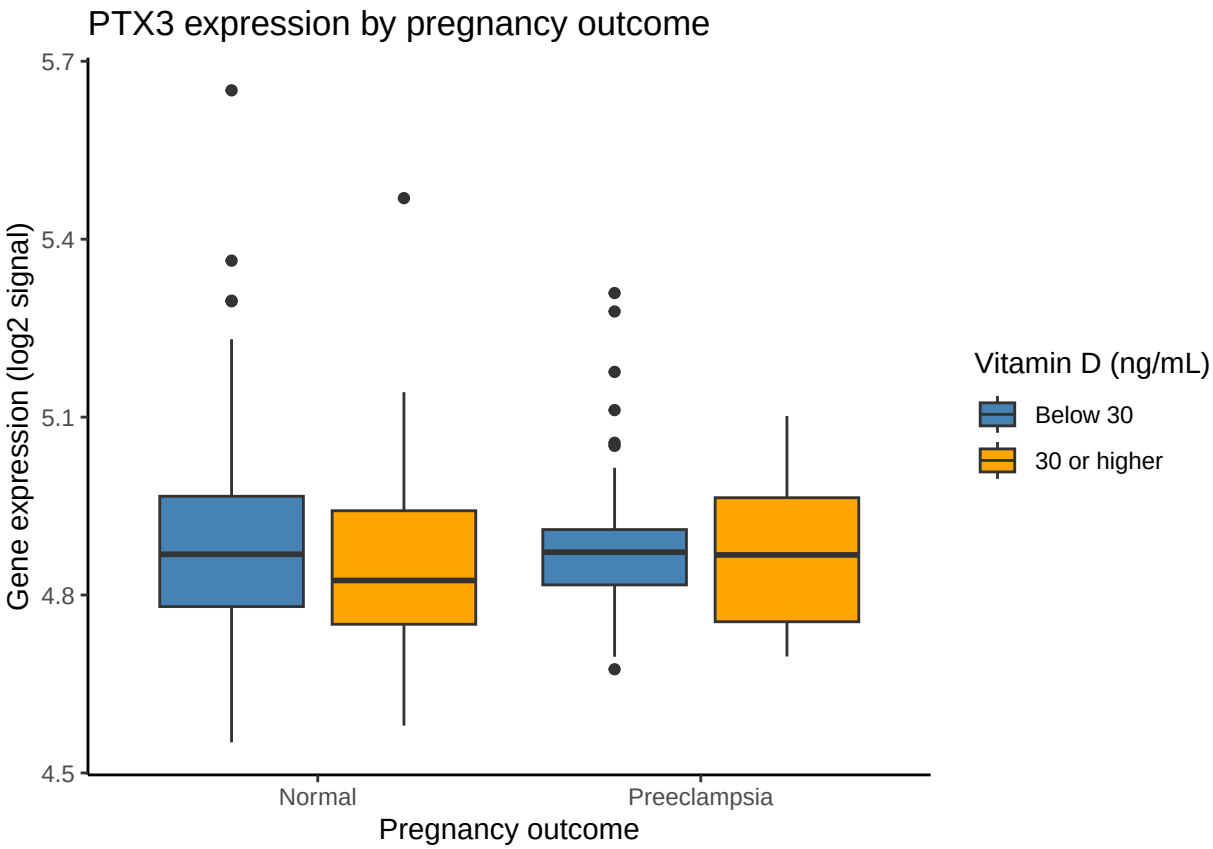

CYP24A1 expression and maternal vitamin D
Gene expression (log2 signal)
5.2
5.0
4.8
4.6
20
40
60
80
Maternal vitamin D (ng/mL)


CYP24A1 expression by pregnancy outcome
Gene expression (log2 signal)
5.2
5.0
4.8
4.6
Normal
Preeclampsia
Pregnancy outcome
Vitamin D (ng/mL)
Below 30
30 or higher




```
## `geom_smooth()` using formula = 'y ~ x'
```


PTX3 expression and maternal vitamin D
Gene expression (log2 signal)
5.7
5.4
5.1
4.8
4.5
20
40
60
80
Maternal vitamin D (ng/mL)


PTX3 expression by pregnancy outcome
Gene expression (log2 signal)
5.7
5.4
5.1
4.8
4.5
Normal
Preeclampsia
Pregnancy outcome
Vitamin D (ng/mL)
Below 30
30 or higher